Part 1: NoSQL Schema Design for EV Population Data

- **Collection Structure (ev_db.vehicles)**:
    - In MongoDB (a document store), each document corresponds to an electric vehicle record from the Washington EV dataset.
    - **Key Fields Indexed & Queried**: Make, Model, Model Year, Electric Vehicle Type, VIN (1-10).
- **Design Rationale**:
    - Storing records as flexible JSON/BSON documents allows fast ingestion of semi-structured CSV data without rigid table constraints.
    - Indexing high-frequency query fields (Make) ensures sub-second aggregation performance during primary and secondary read operations.

Part 2: Tunable Consistency & API Architecture

- **Write Concerns (/insert-fast vs. /insert-safe)**:
    - /insert-fast utilizes WriteConcern(w=1), acknowledging writes as soon as the primary node persists the data. This provides high write throughput and low latency at the cost of durability if the primary crashes before replication.
    - /insert-safe utilizes WriteConcern(w="majority"), requiring confirmation from a majority of replica set nodes. This guarantees strong durability and data safety across network partitions.
- **Read Preferences (/count-tesla-primary vs. /count-bmw-secondary)**:
    - /count-tesla-primary uses ReadPreference.PRIMARY, ensuring strongly consistent reads directly from the primary node to prevent stale data.
    - /count-bmw-secondary uses ReadPreference.SECONDARY_PREFERRED, routing read queries to secondary nodes to offload primary database traffic and improve read scalability, accepting eventual consistency.

Part 3: Performance & Trade-off Analysis

- **Consistency vs. Latency Trade-offs**:
    - Stronger consistency (w="majority" and PRIMARY reads) introduces network round-trip overhead and queueing delays across replica nodes, increasing transaction latency.
    - Weakening consistency (w=1 and SECONDARY_PREFERRED) significantly lowers latency and improves system availability/throughput, but risks reading stale replicas or losing un-replicated writes during failovers.